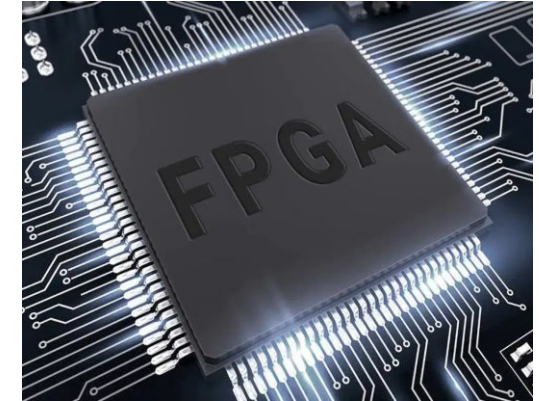


From **Brain-Inspiration** to **Silicon-Realisation**: SNN-based Smart Watchdogs for RISC-V Fault Detection



PhD Student: David Simpson
PhD Supervisor: Prof Jim Harkin
Category: Adaptive Computing
Team Number: AOHW_620

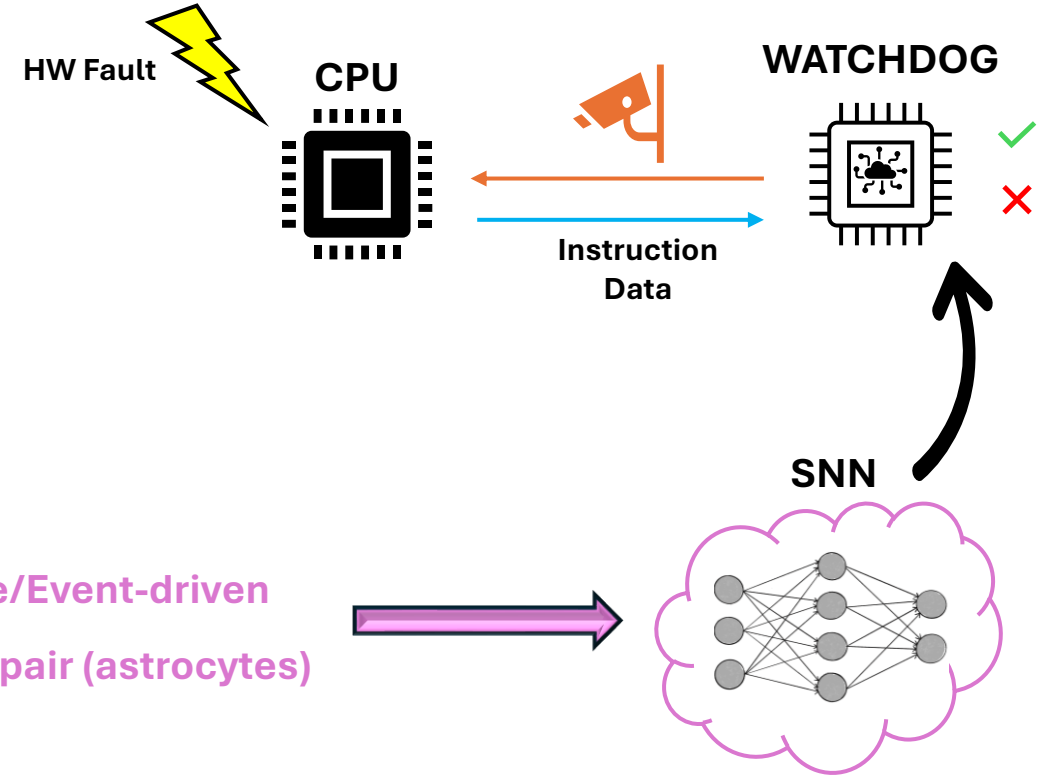


School of Computing, Engineering
and Intelligent Systems



- Modern CPUs can experience hardware faults:

- Radiation / Manufacturing defects
- Reliability of safety-critical applications 
- Watchdogs are required to detect CFEs 

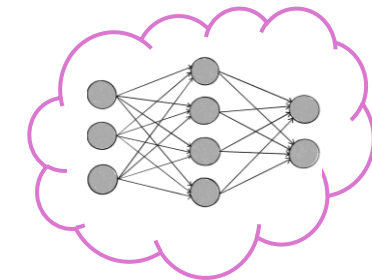


- Watchdog design requirements:

- ✓ Hardware friendly
- ✓ Power efficient
- ✓ Dependable



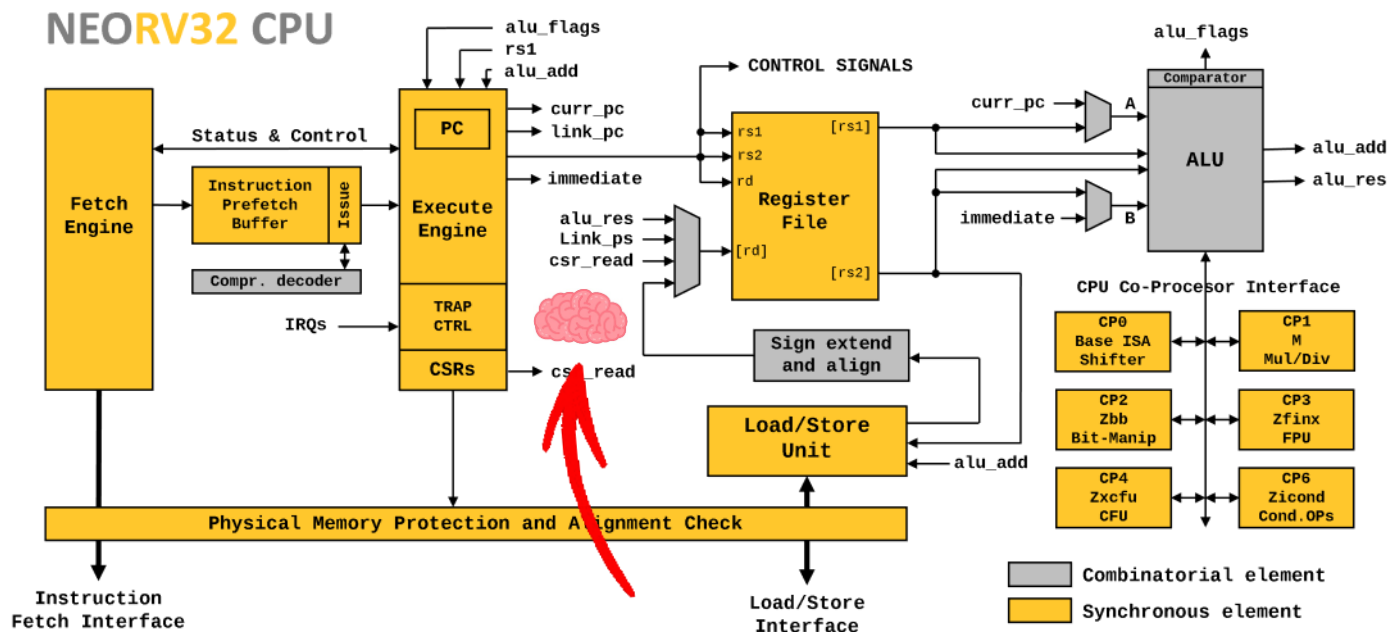
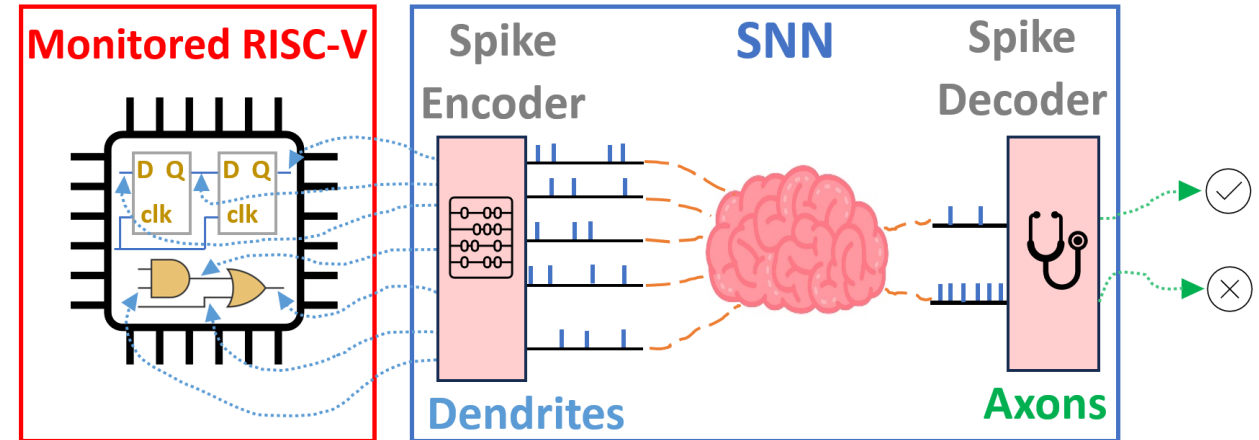
- ❖ Sparse/Event-driven
- ❖ Self repair (astrocytes)



Problem Statement:

- Can a brain-inspired neuromorphic approach using SNNs realise a more efficient and reliable watchdog?

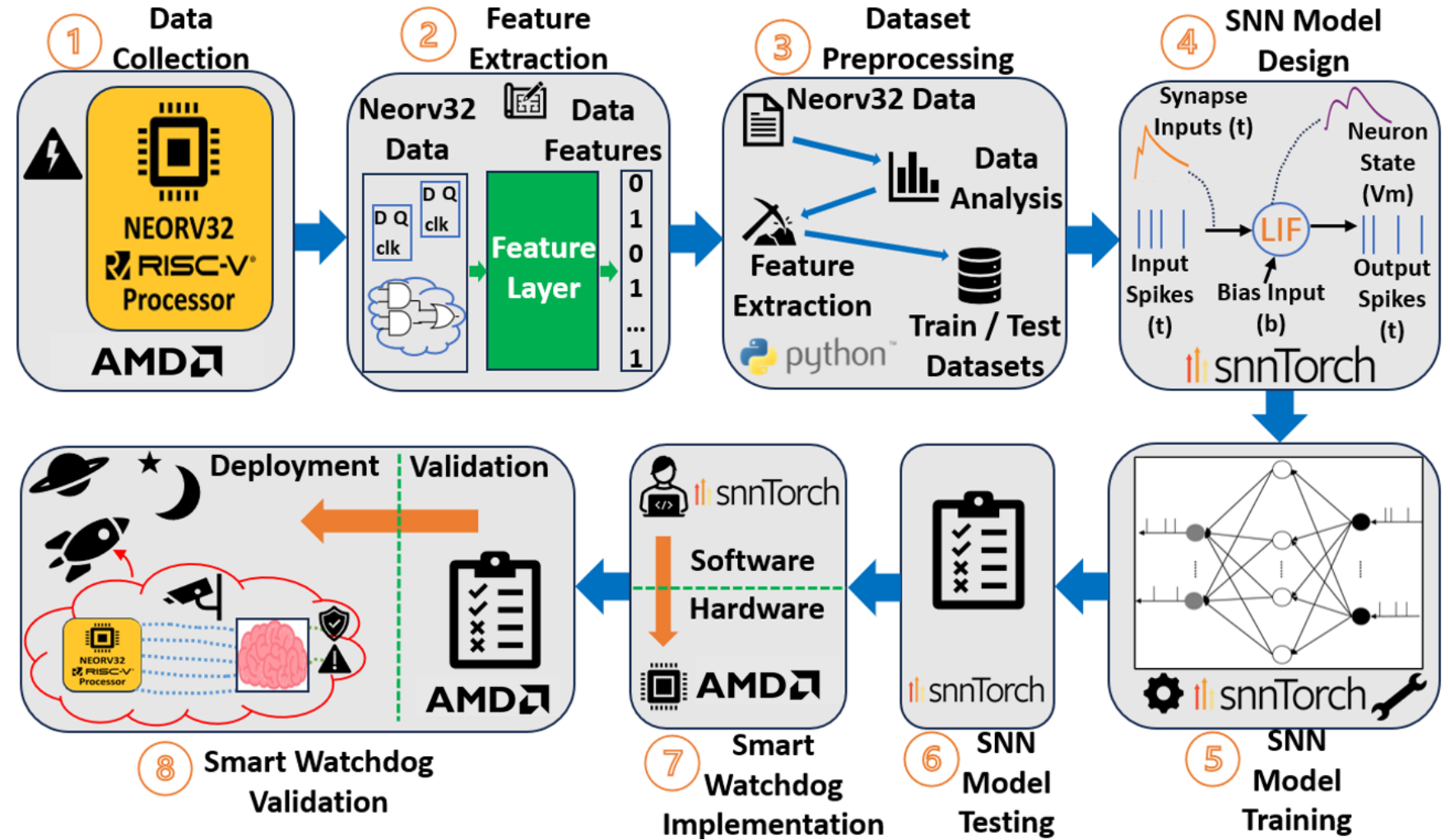
- SNN is the decision-maker in smart watchdog
- 1st SNN model to detect hw faults in CPUs!
- Aim of PhD is to explore fault detection ability



- RISC-V used in this work:
 - Neorv32 software CPU
- Smart Watchdog at Ex pipeline stage

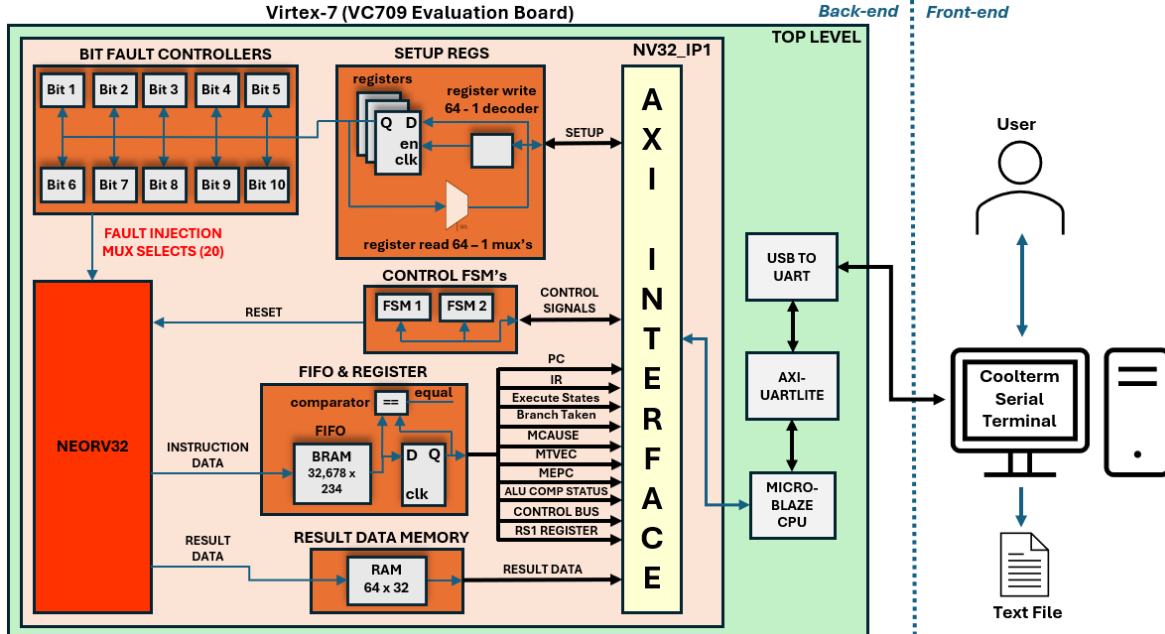
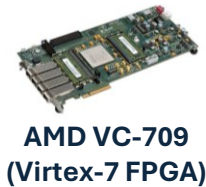
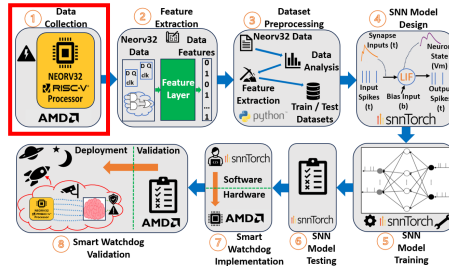
8- Stage Methodology in this PhD

➤ Complete ML Workflow

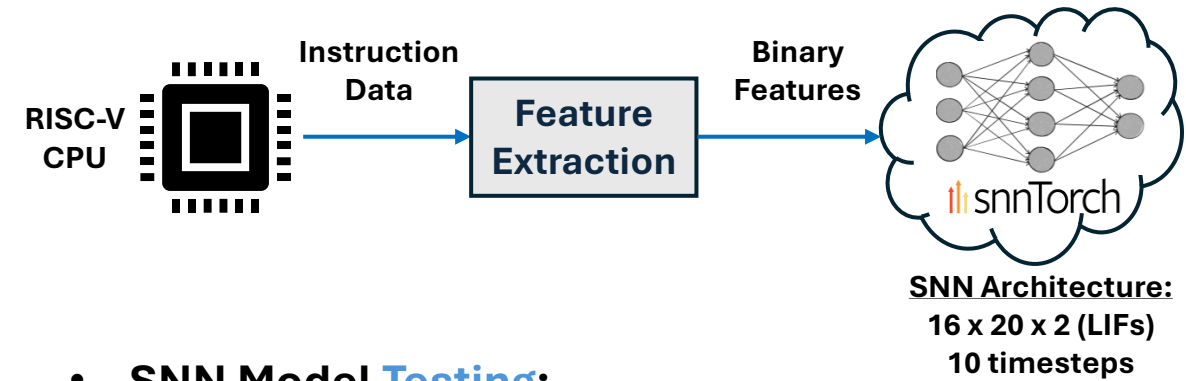
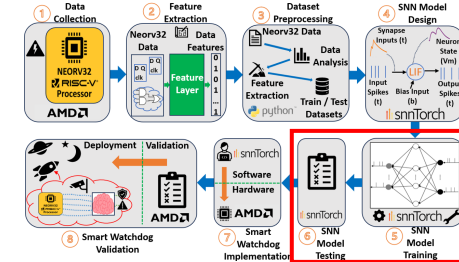


• Data Collection Framework:

- Custom architecture
- Fault Injection / Data Collection
- MicroBlaze / UART / AXI
- Vivado / Vitis IDE



• SNN Model Training:



• SNN Model Testing:

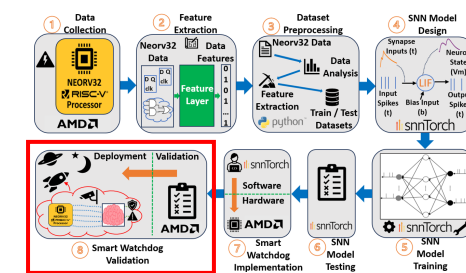
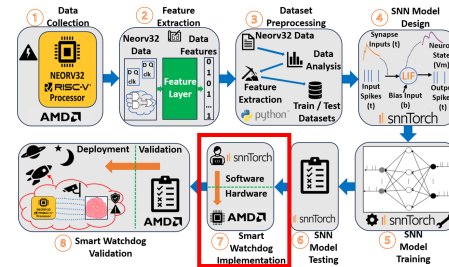
- Unseen C applications

Test Results

Samples	Correct	TP	TN	FN	FP
155	152	78	74	3	0
Accuracy		Precision	Recall	F1 Score	
0.98		1.00	0.96	0.98	

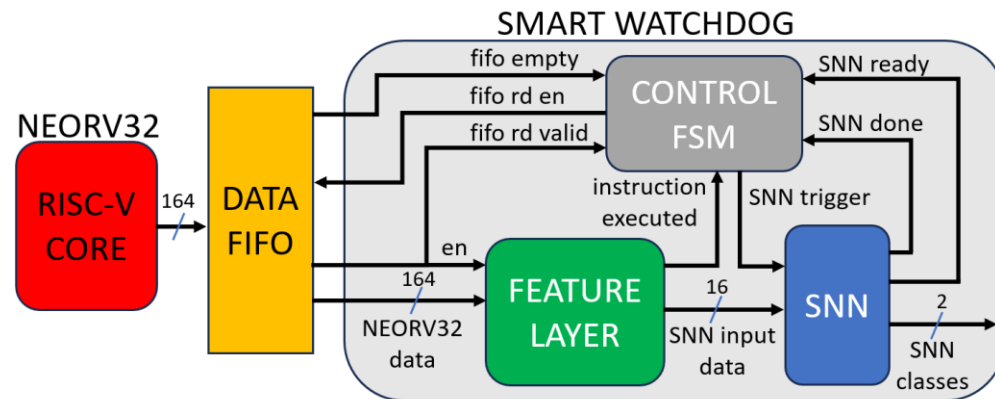


98% accuracy



Hardware Implementation:

- Custom RTL (VHDL)
- In-Silicon (FPGA)



Model sw → hw:



Hardware Validation:

- Unseen C application (Heap Sort)

Validation Results

Samples	Correct	TP	TN	FN	FP
100	98	46	52	2	0
Accuracy		Precision	Recall	F1 Score	
0.98		1.00	0.96	0.98	

✓ 98% accuracy

- Detected silent CFEs (No RISC-V trap) ✓

- [illegible]

Smart Watchdog PPA:

Performance

SNN Model	Clock Cycles	Max Frequency	Inference Latency
Time-mux SNN	~1,433	100MHz ¹	~14.22μs ²
Fast SNN	153	350MHz	~438ns ³

¹ The time-mux SNN was not optimized for timing, therefore max frequency was not determined.

² Latency is therefore calculated at the default frequency of 100MHz.

³ Can be further optimized.

Power and Area

Component	FF's	LUT's	LUTRAM	BRAM	Power (W)
Neorv32 (peripherals)	2,336	2,183	24	8	0.032
Feature Layer	149	157	0	0	0.001
Smart Watchdog (Time-mux SNN)	1,944	2,119	500	0	0.038
Smart Watchdog (Fast SNN)	9,469	6,817	0	0	0.085

Neuromorphic Hardware – not FPGA!



Summary:

- High fault detection capability
- Detects CFEs that RISC-V fails to
- Re-usable with any C code (RV32I)
- Re-usable to other RISC-V cores
- Applicable to other CPU architectures

Future Work:

- Address latency challenges
- Explore temporal spike encoding (efficiency vs noise)
- Develop other smart watchdogs (instruction fetch)
- Astrocyte integration

Power of AMD FPGAs + toolchains in research! 🚀

(Brain-Inspiration to Silicon-Realisation)

Thanks once again to AMD and
Stephan Nolting!

And congrats to all winners! 😊

Contacts:

simpson-d12@ulster.ac.uk

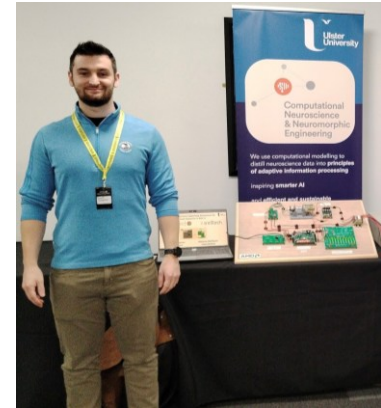
[David Simpson | LinkedIn](#)



Github repo: [DS-567/AMD-AOHW25_620: AMD Open Hardware Design Competition 2025 Contribution: SNN-based Smart Watchdogs for RISC-V Fault Detection](#)

Live Demos

AMD / eFutures
Edge AI 2024



International
Workshop on
Reliable and
Sustainable
Neuromorphic
Hardware 2025



IEEE ISCAS 2025

